

Package ‘CohortPrevalence’

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Title Standardized Prevalence Calculator for OMOP/OHDSI Ecosystem

Version 0.1.0

Description This package calculates prevalence of a condition in a population.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

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Depends R (>= 4.1.0)

Imports cli,
crayon,
fs,
purrr,
SqlRender,
snakecase,
DatabaseConnector,
dplyr,
glue,
readr,
tibble,
tidyr,
here,
methods

Additional_repositories <https://OHDSI.github.io/drat>

Suggests knitr,
rmarkdown,
testthat (>= 3.0.0)

VignetteBuilder knitr

Config/testthat/edition 3

R topics documented:

createCohortPrevalenceAnalysis	2
createDemographicConstraints	3
createDenominatorType	4
createLookBackOptions	4

createPopulationCohort	5
createPrevalenceCohort	5
createRassenIncidenceAnalysis	6
createSpan	7
createTargetCohort	7
createYearlyRange	8
exportPrevalenceQuery	8
exportPrevalenceResults	9
generateMultiplePrevalence	9
generateMultipleRassenIncidence	10
generateSinglePrevalence	10
generateSingleRassenIncidence	11

Index	12
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createCohortPrevalenceAnalysis

Create a CohortPrevalenceAnalysis object

Description

Constructs an CohortPrevalenceAnalysis object with the specified settings.

Usage

```
createCohortPrevalenceAnalysis(
  analysisId,
  prevalentCohort,
  periodOfInterest,
  lookBackOptions,
  numeratorType,
  denominatorType,
  minimumObservationLength = 0L,
  useOnlyFirstObservationPeriod = FALSE,
  multiplier = 100000L,
  strata = NULL,
  demographicConstraints = createDemographicConstraints(),
  populationCohort = NULL
)
```

Arguments

analysisId	Unique integer analysisId to identify the analysis (required).
prevalentCohort	A PrevalenceCohort object specifying the cohort of interest (required).
periodOfInterest	A PeriodOfInterest object (required).
numeratorType	Character string specifying numerator type. Must be one of: "pn1": Patients who have been observed to have the condition of interest on the first day of the period of interest or within the lookback time

- "pn2": patients who have been observed to have the condition of interest at any time in the period of interest or within the lookback time

denominatorType	A DenominatorType object (required).
useOnlyFirstObservationPeriod	Logical: TRUE to restrict analysis to the first observation period (optional).
multiplier	Integer specifying prevalence multiplier (optional).
strata	Character string. Must be one, or some of: "age", "gender", "race" (optional).
demographicConstraints	a DemoConstraint object specifying the constraints of the population.
populationCohort	A CohortPopulation object specifying the population of interest on which to compute prevalence.
lookbackOptions	A LookBackOption object (required).
minimumObservationLength:	Integer specifying minimum observation length (optional).

Value

A CohortPrevalenceAnalysis R6 object.

createDemographicConstraints
Create a DemoConstraint object

Description

Constructs an DemoConstraint object for prevalence analyses.

Usage

```
createDemographicConstraints(
  ageMin = 0,
  ageMax = 150,
  genderIds = c(8507, 8532)
)
```

Arguments

ageMin	The minimum age allowed for the population. Default is 0
ageMax	the maximum age allowed for the population. Default is 150
genderIds	the genderIds allowed. Default is 8507 - M, and 8532 - F

Value

A DemoConstraint R6 object.

`createDenominatorType` *Create a DenominatorType object*

Description

Constructs an `DenominatorType` object for denominator choice.

Usage

```
createDenominatorType(denomType, sufficientDays = NULL)
```

Arguments

`denomType` Character string specifying denominator type. Must be one of:

- "pd1": Patients who have been observed on the first day of the period of interest
- "pd2": Patients who contribute all observable person-days in the period of interest.
- "pd3": Patients who contribute at least 1 day in the period of interest.
- "pd4": Patients who contribute sufficient time in the period of interest based on at least n observable person-days in the period of interest.

`sufficientDays` Integer: For denominator choice "pd4", the number of minimum observable days patients must be observed.

Value

A `DenominatorType` R6 object.

`createLookBackOptions` *Create a LookBackOptions object*

Description

Constructs an `LookBackOptions` object with the specified settings.

Usage

```
createLookBackOptions(lookBackDays = 99999L, useObservedTimeOnly = FALSE)
```

Arguments

`lookBackDays` An integer number of days for the lookback period.

`useObservedTimeOnly` Logical: TRUE restricts the lookback period to only using observed periods.

Value

A `LookBackOptions` R6 object.

`createPopulationCohort`*Create a population cohort CohortInfo object*

Description

Constructs an CohortInfo object for population of interest.

Usage

```
createPopulationCohort(cohortId, cohortName)
```

Arguments

<code>cohortId</code>	Integer: the cohort ID within the database results schema of interest.
<code>cohortName</code>	Character string specifying a name for the cohort.

Value

A CohortInfo R6 object.

`createPrevalenceCohort`*Create a prevalence cohort CohortInfo object*

Description

Constructs an CohortInfo object for target cohort of interest

Usage

```
createPrevalenceCohort(cohortId, cohortName)
```

Arguments

<code>cohortId</code>	Integer: the cohort ID within the database results schema of interest.
<code>cohortName</code>	Character string specifying a name for the cohort.

Value

A CohortInfo R6 object.

```
createRassenIncidenceAnalysis
```

Create a IncidenceAnalysis object for Rassen Incidence

Description

Constructs an IncidenceAnalysis object with the specified settings.

Usage

```
createRassenIncidenceAnalysis(
  analysisId,
  targetCohort,
  periodOfInterest,
  minimumObservationLength = 0L,
  useOnlyFirstObservationPeriod = FALSE,
  multiplier = 100000L,
  strata = NULL,
  demographicConstraints = createDemographicConstraints(),
  populationCohort = NULL
)
```

Arguments

analysisId	Unique integer analysisId to identify the analysis (required).
targetCohort	A TargetCohort object specifying the cohort of interest (required).
periodOfInterest	A PeriodOfInterest object (required).
useOnlyFirstObservationPeriod	Logical: TRUE to restrict analysis to the first observation period (optional).
multiplier	Integer specifying prevalence multiplier (optional).
strata	Character string. Must be one, or some of: "age", "gender", "race" (optional).
demographicConstraints	a DemoConstraint object specifying the constraints of the population.
populationCohort	A CohortPopulation object specifying the population of interest on which to compute prevalence.
minimumObservationLength:	Integer specifying minimum observation length (optional).

Value

A IncidenceAnalysis R6 object.

createSpan	<i>Create a PeriodOfInterest object</i>
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Description

Constructs an PeriodOfInterest object for span prevalence analyses.

Usage

```
createSpan(startDates, endDates)
```

Arguments

startYears	A numeric vector of start years of interest.
endYears	A numeric vector of end years of interest.

Value

A PeriodOfInterest R6 object.

createTargetCohort	<i>Create a target cohort CohortInfo object</i>
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Description

Constructs an CohortInfo object for target cohort of interest

Usage

```
createTargetCohort(cohortId, cohortName)
```

Arguments

cohortId	Integer: the cohort ID within the database results schema of interest.
cohortName	Character string specifying a name for the cohort.

Value

A CohortInfo R6 object.

createYearlyRange	<i>Create a PeriodOfInterest object</i>
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Description

Constructs an PeriodOfInterest object for yearly prevalence analyses.

Usage

```
createYearlyRange(range)
```

Arguments

range	A numeric vector of years of interest.
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Value

A PeriodOfInterest R6 object.

exportPrevalenceQuery	<i>Export Prevalence Query</i>
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Description

Exports the full SQL query of a CohortPrevalenceAnalysis analysis.

Usage

```
exportPrevalenceQuery(
  prevalenceAnalysisClass,
  executionSettings,
  outputFolder = NULL
)
```

Arguments

prevalenceAnalysisClass	A CohortPrevalenceAnalysis R6 object with analysis settings (required).
outputFolder	Character string specifying the path to the folder where the output files will be saved. If left NULL, will default to current working directory (optional).

`exportPrevalenceResults`*Export Prevalence Query*

Description

Saves the results of a CohortPrevalenceAnalysis analysis as a .csv.

Usage

```
exportPrevalenceResults(results, outputFolder = NULL)
```

Arguments

<code>results</code>	Dataframe: Result of a generateSinglePrevalence analysis.
<code>outputFolder</code>	Character string specifying the path to the folder where the output files will be saved. If left NULL, will default to current working directory (optional).

`generateMultiplePrevalence`*Run Multiple Prevalence Analyses*

Description

Runs multiple prevalence analysis with a list of specified CohortPrevalenceAnalysis settings

Usage

```
generateMultiplePrevalence(prevalenceAnalysisList, executionSettings)
```

Arguments

<code>prevalenceAnalysisList</code>	A list CohortPrevalenceAnalysis R6 object with analysis settings.
<code>executionSettings</code>	An executionSettings R6 object with connection and schema details.

Value

A results dataframe with prevalence rates and strata per analysis id.

`generateMultipleRassenIncidence`*Run Multiple Rassen Incidence Analyses*

Description

Runs multiple incidence analysis with a list of specified IncidenceAnalysis settings

Usage

```
generateMultipleRassenIncidence(incidenceAnalysisList, executionSettings)
```

Arguments

`incidenceAnalysisList`

A list IncidenceAnalysis R6 object with analysis settings.

`executionSettings`

An executionSettings R6 object with connection and schema details.

Value

A results dataframe with incidence rates and strata per analysis id.

`generateSinglePrevalence`*Run Single Prevalence Analysis*

Description

Runs a single prevalence analysis with specified CohortPrevalenceAnalysis settings

Usage

```
generateSinglePrevalence(prevalenceAnalysisClass, executionSettings)
```

Arguments

`prevalenceAnalysisClass`

A CohortPrevalenceAnalysis R6 object with analysis settings.

`executionSettings`

An executionSettings R6 object with connection and schema details.

Value

A results dataframe with prevalence rates and strata.

`generateSingleRassenIncidence`*Run Single Incidence Analysis*

Description

Runs a single incidence analysis with specified IncidenceAnalysis settings

Usage

```
generateSingleRassenIncidence(incidenceAnalysisClass, executionSettings)
```

Arguments

`incidenceAnalysisClass`

A IncidenceAnalysis R6 object with analysis settings.

`executionSettings`

An executionSettings R6 object with connection and schema details.

Value

A results dataframe with incidence rates and strata.

Index

`createCohortPrevalenceAnalysis`, [2](#)
`createDemographicConstraints`, [3](#)
`createDenominatorType`, [4](#)
`createLookBackOptions`, [4](#)
`createPopulationCohort`, [5](#)
`createPrevalenceCohort`, [5](#)
`createRassenIncidenceAnalysis`, [6](#)
`createSpan`, [7](#)
`createTargetCohort`, [7](#)
`createYearlyRange`, [8](#)

`exportPrevalenceQuery`, [8](#)
`exportPrevalenceResults`, [9](#)

`generateMultiplePrevalence`, [9](#)
`generateMultipleRassenIncidence`, [10](#)
`generateSinglePrevalence`, [10](#)
`generateSingleRassenIncidence`, [11](#)